## Question 1

**1. Load the data into a variable or dataset (in the form of a variable frame).**

```
data_1 = pd.read_csv('/content/data_1.csv')
data_1.head(5)
```

**2. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.info()
data_1.isnull().sum()
data_1.shape
data_1.describe(include='object')
data_1.describe(include='float64')
data_1.describe(include='int64')
```

```
<class>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 21):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Unnamed: 0              1000 non-null   int64
 1   Age                     1000 non-null   int64
 2   Sex                     1000 non-null   object
 3   Job                     1000 non-null   int64
 4   Housing                 1000 non-null   object
 5   Saving accounts         817 non-null    object
 6   Checking account        606 non-null    object
 7   Credit amount           1000 non-null   int64
 8   Duration                1000 non-null   int64
 9   Purpose                 1000 non-null   object
 10  Risk                    1000 non-null   object
 11  Unnamed: 0.1            1000 non-null   int64
 12  Unnamed: 0.2            1000 non-null   int64
 13  Unnamed: 0.3            1000 non-null   int64
 14  Unnamed: 0.4            1000 non-null   int64
 15  Unnamed: 0.5            1000 non-null   int64
 16  Unnamed: 0.6            1000 non-null   int64
 17  Unnamed: 0.7            1000 non-null   int64
 18  Unnamed: 0.8            1000 non-null   int64
 19  Unnamed: 0.9            1000 non-null   int64
 20  Unnamed: 0.10           1000 non-null   int64
dtype: int64
(1000, 21)
```

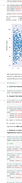
```
Unnamed: 0  Age  Sex  Job  Housing  Saving accounts  Checking account  Credit amount  Duration  Purpose  Risk  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.0  1000.0  1000  1000.0  1000  817  606  1000.0  1000.0  1000  1000  1000.0  1000.0  1000.0  1000.0  1000.0  1000.0  1000.0  1000.0  1000.0  1000.0
unique  NaN  NaN  2  NaN  3  4  3  NaN  NaN  8  2  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN
top  NaN  NaN  male  NaN  own  little  little  NaN  NaN  car  good  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN
freq  NaN  NaN  690  NaN  713  183  274  NaN  NaN  337  700  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN
```

```
Unnamed: 0  Age  Job  Credit amount  Duration  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  35.546000  1.904000  3271.258000  20.903000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  11.375469  0.653448  2822.736876  12.058814  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  19.000000  0.000000  250.000000  4.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  27.000000  2.000000  1365.500000  12.000000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  33.000000  2.000000  2319.500000  18.000000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  42.000000  2.000000  3972.250000  24.000000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  75.000000  3.000000  18424.000000  72.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**3. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.isnull()
data_1.isnull().sum()
```

```
Unnamed: 0              0
Age                     0
Sex                     0
Job                     0
Housing                 0
Saving accounts       183
Checking account      394
Credit amount           0
Duration                0
Purpose                 0
Risk                    0
Unnamed: 0.1            0
Unnamed: 0.2            0
Unnamed: 0.3            0
Unnamed: 0.4            0
Unnamed: 0.5            0
Unnamed: 0.6            0
Unnamed: 0.7            0
Unnamed: 0.8            0
Unnamed: 0.9            0
Unnamed: 0.10           0
dtype: int64
```




Do we need to drop the columns that contain missing values? Why or why not? Yes, we need to drop the columns that contain missing values because the data is not complete and it may affect the analysis. Missing values can lead to biased results and reduce the accuracy of the model. In this case, the columns "Saving accounts" and "Checking account" have a significant number of missing values, which could potentially skew the results. Therefore, it is better to drop these columns to ensure the integrity of the analysis.

**4. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.drop(['Saving accounts', 'Checking account'], axis=1, inplace=True)
data_1.isnull().sum()
```

```
Unnamed: 0              0
Age                     0
Sex                     0
Job                     0
Housing                 0
Credit amount           0
Duration                0
Purpose                 0
Risk                    0
Unnamed: 0.1            0
Unnamed: 0.2            0
Unnamed: 0.3            0
Unnamed: 0.4            0
Unnamed: 0.5            0
Unnamed: 0.6            0
Unnamed: 0.7            0
Unnamed: 0.8            0
Unnamed: 0.9            0
Unnamed: 0.10           0
dtype: int64
```

**5. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.drop(['Unnamed: 0', 'Unnamed: 0.1', 'Unnamed: 0.2', 'Unnamed: 0.3', 'Unnamed: 0.4', 'Unnamed: 0.5', 'Unnamed: 0.6', 'Unnamed: 0.7', 'Unnamed: 0.8', 'Unnamed: 0.9', 'Unnamed: 0.10'], axis=1, inplace=True)
data_1.isnull().sum
```

```
Age                 0
Sex                 0
Job                 0
Housing             0
Credit amount       0
Duration            0
Purpose             0
Risk                0
dtype: int64
```

**6. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.shape
```

```
(1000, 8)
```

**7. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**8. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**9. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**10. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**11. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**12. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**13. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**14. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**15. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**16. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**17. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**18. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**19. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**20. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**21. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**22. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**23. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**24. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**25. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**26. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**27. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**28. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**29. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**30. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**31. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**32. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**33. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**34. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**35. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**36. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**37. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**38. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**39. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000  999.000000
```

**40. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='object')
```

```
Sex  Housing  Purpose  Risk
count  1000  1000  1000  1000
unique  2  3  8  2
top  male  own  car  good
freq  690  713  337  700
```

**41. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='int64')
```

```
Age  Job  Credit amount  Duration
count  1000.000000  1000.000000  1000.000000  1000.000000
mean  35.546000  1.904000  3271.258000  20.903000
std  11.375469  0.653448  2822.736876  12.058814
min  19.000000  0.000000  250.000000  4.000000
25%  27.000000  2.000000  1365.500000  12.000000
50%  33.000000  2.000000  2319.500000  18.000000
75%  42.000000  2.000000  3972.250000  24.000000
max  75.000000  3.000000  18424.000000  72.000000
```

**42. Identify the data. Was the variable or dataset in the dataset the same after variable or dataset? Was the dataset same as missing data?**

```
data_1.describe(include='float64')
```

```
Unnamed: 0  Unnamed: 0.1  Unnamed: 0.2  Unnamed: 0.3  Unnamed: 0.4  Unnamed: 0.5  Unnamed: 0.6  Unnamed: 0.7  Unnamed: 0.8  Unnamed: 0.9  Unnamed: 0.10
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000  1000.000000
mean  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
std  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436  288.819436
min  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.000000
25%  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000  249.750000
50%  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000  499.500000
75%  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000  749.250000
max  999.000000  999.000000
```